2026 START Program

Anticipate, Simulate, Replan - Egocentric World Models for Multi-Step Agentic Reasoning

PART 1: Proposal Identification

START Theme	2.0 Agentic, Artificial Intelligence
Sub-theme	2.1 Multimodal Reasoning Agents for Complex, Long-Horizon Tasks
Proposal Title	Anticipate, Simulate, Replan - Egocentric World Models for Multi-Step Agentic Reasoning

PI & Co PI information

Full Name	Lorenzo Torresani
Name of Department	Khoury College of Computer Sciences
Name of Institute	Northeastern University
Contact Email	L.Torresani@northeastern.edu
Contact Phone Number	650-690-0070

Statement of Joint Proposal: N/A

PART 2: Project Summary

Executive summary

We propose to develop multimodal agents that can anticipate, simulate, and replan during long-horizon tasks from egocentric observations. Our central hypothesis is that reliable multi-step reasoning requires tight coupling between semantic future simulation and grounded plan generation. We will introduce (i) a latent egocentric world model that predicts future scene representations conditioned on ego-motion trajectories and language-grounded semantic targets, and (ii) a planning policy trained to generate grounded action sequences and adapt them online via model-based simulation. Validated on large-scale egocentric benchmarks and an end-to-end embodied demonstration, this work will deliver open-source models, a closed-loop evaluation protocol, and multiple top-tier publications establishing new state of the art in multimodal agentic reasoning. We expect the resulting system to enable robust assistance in real-world and device-centric tasks, including procedural guidance, UI workflows, and error recovery, directly aligning with next-generation Galaxy AI agents.

Response to Reviewer Feedback

This proposal preserves the architectural framing, scope, team composition, and core technical commitments of our preliminary submission. We sharpen the technical specification of each component in response to the three explicit asks in the reviewers' feedback.

Differentiation from latest world model works (LeWorldModel, AdaWorld, etc.). Addressed in the *Differentiation from Recent Latent World Model Work* subsection. We compare against LeWorldModel [22] and AdaWorld [23] (named in the feedback) and additionally against V-JEPA 2 / V-JEPA 2-AC [2] as the reference precedent in the JEPA-for-robotics line, and EgoAgent [25] as the closest existing egocentric joint predictive agent.

End-to-end demonstration on egocentric robotic manipulation or navigation, with task success improvement. Addressed in the *End-to-End Demonstration Plan* subsection. We propose the Ego Humanoid Manipulation Benchmark [21] as the end-to-end embodied demonstration, evaluated against the EgoVLA and no-pretraining published baselines on both seen and unseen visual backgrounds.

Details of benchmarks (simulation) targeted for VLA or VLN. The Ego Humanoid Manipulation Benchmark is our VLA evaluation target. The preliminary proposal's procedural-video evaluation suite (EgoPER [16], ExAct [17], COIN [18], CrossTask [19], Ego4D-v2 [20]) is preserved as the world-model diagnostic surface in its native domain.

What is unchanged from the preliminary proposal. The three core architectural components (EgoLWM world model, GGPO planner, integrated Anticipate–Simulate–Replan Loop), the 12-month duration, the team composition, the budget, the Year 2/3 vision (digital/UI bridge and on-device deployment), and the open-source release commitments are unchanged. What changes is the sharpened technical specification of EgoLWM (an action-aligned latent target as an extension of LeWM's [22] training framework, with hand-pose readout supervision aligned to the embodied benchmark's action space), and the addition of the end-to-end embodied demonstration that the preliminary proposal did not specify.

Research Abstract & Goals

What problem are we trying to solve?
What if an AI assistant could watch you assemble flat-pack furniture, anticipate that a bracket being installed will cause a structural failure later, and replan the remaining steps before you reach the point of no return? Realizing this vision requires agents that reason causally within the world, anticipating how actions will unfold, simulating what could go wrong, and replanning adaptively when they do. EgoLWM is the engine that makes anticipation and simulation possible, giving an agent the ability to look ahead in semantic latent space and ask whether continuing on the current path will lead to failure. GGPO is the engine that makes adaptive planning possible, teaching an agent to generate and revise multi-step plans grounded in how humans actually execute tasks. Current multimodal models fail at precisely this [24]. This gap is particularly critical for device-based assistants, where users expect continuous, multi-step guidance across physical and digital environments. See Figure 1 for illustration.

What approach or method are you going to take?
Our approach is based on the principle that **prediction and planning must be learned jointly**, rather than as independent modules.

Egocentric Latent World Model (EgoLWM). We design a world model that predicts how a first-person scene will evolve, conditioned on 6DOF camera ego-motion [1] as the per-step disambiguating signal, with language-grounded action embeddings serving as a training-time anchor that shapes an action-aligned latent space. Critically, EgoLWM operates in semantic latent space rather than at the pixel level [3, 4], making it faster and directly aligned with planning tasks. The preliminary proposal committed to a frozen V-JEPA 2 encoder [2]; building on diagnostic studies described in the Technical Approach, we sharpen this to an end-to-end trained encoder predicting in an action-aligned semantic latent space, extending the stable-training framework of LeWM [22]. Ego-motion is injected globally via adaptive layer normalization, and a trajectory dropout mechanism enables graceful degradation when trajectory estimation is unreliable. EgoLWM is evaluated on the preliminary proposal's procedural-video suite (mistake anticipation on EgoPER [16], proficiency on ExAct [17], step forecasting on COIN [18] and CrossTask [19], long-horizon anticipation on Ego4D-v2 [20]) and on the Ego Humanoid Manipulation Benchmark [21] for end-to-end embodied demonstration.

Grounding-Guided Planning Policy (GGPO). Rather than extracting noisy pseudo-labels from instructional video, GGPO uses large-scale video as a plausibility oracle: a generated step sequence earns RL reward [7] proportional to how well it grounds in real ASR narrations [5], a verification problem rather than a labeling problem. Because multiple valid execution orderings all ground successfully, the policy learns a distribution of valid plans rather than a single trajectory, enabling adaptive replanning. Training combines clean-label supervision [15, 12] with HowTo100M-scale [5] weakly-labeled video through the grounding-based reward, scaling the policy across data regimes.

Integrated Anticipate–Simulate–Replan Loop. The system operates in a closed loop (see Figure 2): GGPO proposes candidate action sequences (e.g., key steps or waypoints), and EgoLWM simulates their future egocentric latent trajectories conditioned on the goal. The agent selects and executes actions based on simulated outcomes, while continuously comparing predicted and observed latent trajectories; discrepancies trigger GGPO to revise its plan online. This continual interplay between simulation and action enables implicit replanning, allowing the agent to recover from errors and adapt to changing conditions while maintaining coherent long-horizon behavior. Because planning informs simulation and simulation informs planning, the system anticipates failures before they occur. Recent empirical work [24] identifies precisely this integration as a key bottleneck for current agentic systems, which the proposed closed-loop design directly addresses.

Background and significance

Both contributions build on the PI's established program: Ego-Exo4D [8] provides egocentric video, 6DOF poses, and proficiency labels; Video ReCap [9], BIMBA [10], and PerceptionLM [11] provide video-language modeling foundations; HT-Step [12], Ego4D Goal-Step [13], and Step Differences [14] underpin GGPO. The proposed capabilities are domain-general: any agent operating in a partially observable environment over extended time horizons requires precisely these components.

What benefit can Samsung expect?

This project enables multimodal agents that provide continuous, multi-step assistance across Samsung devices. EgoLWM's latent-space prediction is orders of magnitude faster than pixel-level diffusion, enabling real-time replanning on mobile and wearable hardware. Recent Samsung work on memory-constrained streaming video understanding [26] has addressed the perception-side bottleneck for hour-long egocentric video on AR glasses and edge robots; the proposed world model is designed to operate downstream of such compressed streaming-video representations, completing the on-device perception-anticipation-action stack for next-generation Galaxy AI assistants. GGPO's weak-supervision approach removes annotation bottlenecks, allowing policies to generalize to new domains without costly labeling. EgoLWM operates on camera trajectories estimated from the video feed alone (no specialized IMU or SLAM hardware required) [1], and the trajectory dropout mechanism further ensures robustness across Samsung's full device range, directly underpinning next-generation Galaxy AI assistants that detect errors, simulate consequences, and adapt plans in real time.

Differentiation from Recent Latent World Model Work

Recent work has rapidly advanced latent world models for embodied prediction and control. We position the proposed approach against four representative efforts: LeWorldModel (LeWM) [22] and AdaWorld [23], named in the reviewers' feedback; V-JEPA 2 / V-JEPA 2-AC [2], the reference precedent in the JEPA-for-robotics line; and EgoAgent [25], the closest prior egocentric joint predictive agent. Empirical evidence that current agents struggle to integrate world-model foresight into action [24] further motivates the closed-loop design we propose.

LeWM establishes a foundational result for the field: a latent world model trained end-to-end from pixels under a two-term objective (predicting the next latent embedding, plus a regularizer that pushes the latents toward an isotropic Gaussian distribution), trained on a single GPU. We propose to build directly on this foundation while re-targeting the latent objective. LeWM optimizes for visual continuity under an isotropic prior, evaluated on third-person 2D and 3D toy control benchmarks. Preliminary diagnostic studies by our group find that visual-continuity-organized latents are poorly suited to action-level discrimination at long horizons: simple distance-based CEM in V-JEPA-style latent space underperforms a no-trajectory baseline at every horizon tested. EgoLWM replaces the isotropic-Gaussian target with an action-aligned target while preserving SIGReg-style anti-collapse, extending LeWM's framework from third-person toy control to first-person multi-step manipulation. We note that

LeWM is co-authored by Damien Scieur (Samsung SAIL); our work is intended as a substantive continuation of an effort with internal Samsung roots, and we would welcome the opportunity to coordinate with the LeWM team where productive.

AdaWorld pretrains a generative diffusion world model (built on Stable Video Diffusion) controlled by a continuous latent action extracted β-VAE-style from consecutive video frames, enabling zero-shot action transfer and efficient adaptation. Our design differs in three central ways. First, we adopt a latent-predictive (JEPA-style) rather than pixel-generative architecture, motivated by AdaWorld's own acknowledged limitations (non-real-time inference, no long-horizon rollouts) that make it unsuitable for the many-iteration Anticipate–Simulate–Replan loop we propose. Second, our action interface operates at the level of semantic subgoals (language-described intermediate steps), not low-level frame-to-frame transitions. Third, AdaWorld evaluates on Procgen arcade games and adaptation to Habitat / Minecraft / DMLab; none address egocentric fine-grained manipulation, the regime we target.

V-JEPA 2 / V-JEPA 2-AC demonstrates that a self-supervised video encoder, frozen and combined with an action-conditioned predictor post-trained on modest robot data, can drive zero-shot pick-and-place at 65–80% success on two Franka arms. This sets a reference for what frozen V-JEPA features support. Our hypothesis, supported by the diagnostic studies described in the Technical Approach, is that a frozen V-JEPA feature space optimized for visual continuity leaves substantial action-discriminative signal unexposed. Rather than freezing the encoder and projecting after the fact, we propose to train the encoder end-to-end against an action-aligned target, accepting the cost of action supervision in exchange for a latent space whose geometry is structurally aligned with action.

EgoAgent is also a JEPA-style egocentric predictive agent, but with a different aim and training regime. EgoAgent jointly learns to perceive, predict, and act in a single transformer, trained on Nymeria (synchronized egocentric video and 3D body motion capture); its "action" is 3D body motion prediction, and it is evaluated on representation quality, future state prediction, and 3D human motion. The proposed system instead generates and revises language-described multi-step plans conditioned on an explicit task goal, with supervision derived from grounding against HowTo100M-scale narrated video (no motion capture required). Architecturally, EgoAgent produces a single behavioral continuation per timestep; the proposed system explicitly simulates candidate plans, scores them against the goal, and revises on observed-versus-predicted surprise.

Description of Project

Project Timeline

a) Project Duration: 09/01/2026 - 08/31/2027
b) Project Scope (year 1):

Quarter	Milestones & Deliverables
Q1 Sep–Nov 2026	Design and validate the LeWM-derived hybrid regularizer that retains SIGReg-style anti-collapse on directions orthogonal to the action manifold while pulling latents toward action-aligned targets; this is the principal new technical contribution distinguishing EgoLWM from the isotropic-Gaussian setting LeWM is constructed for. Build the end-to-end encoder training pipeline on Ego-Exo4D [8] under this regularizer. Implement AdaLN trajectory conditioning. Begin design of the closed-loop integration interface between EgoLWM and GGPO. Q1 gate: the hybrid regularizer produces stable end-to-end training and a latent geometry measurably more action-aligned than V-JEPA-style isotropic targets.
Q2 Dec 2026–Feb 2027	Train EgoLWM with the hybrid regularizer under architectural ablations: AdaLN versus additive trajectory injection; end-to-end versus frozen encoder. Develop the inference-time trajectory predictor that closes the gap between oracle and inference settings. Evaluate the resulting model on procedural-video proficiency benchmarks (ExAct [17]) under the new training framework, characterizing what the regularizer change buys beyond visual-continuity baselines. Scale GGPO to HowTo100M-scale [5] weak supervision via grounding-based reward. Q2 gate: end-to-end trained EgoLWM with hybrid regularizer matches or exceeds frozen-encoder

Quarter	Milestones & Deliverables
	baselines on procedural-video evaluation, with ablations clearly isolating each component's contribution.
Q3 Mar–May 2027	EgoLWM results on procedural planning and long-horizon anticipation benchmarks (COIN [18], CrossTask [19], Ego4D-v2 [20]); embodied evaluation on the Ego Humanoid Manipulation Benchmark [21] begins (short-horizon tasks first); reactive variant (GGPO planner alone with EgoLWM as MANO policy). Q3 gate: full system runs end-to-end on the Ego Humanoid Manipulation Benchmark; reactive variant is competitive with EgoVLA on short-horizon tasks.
Q4 Jun–Aug 2027	Closed-loop integration of EgoLWM and GGPO; full Anticipate–Simulate–Replan evaluation on long-horizon multi-stage tasks (seen and unseen backgrounds); release of closed-loop evaluation protocol; manuscript preparation and submission. Q4 gate: closed-loop system measurably outperforms the reactive variant on long-horizon tasks on at least one of seen and unseen visual backgrounds.

c) Overall Project Plan:
Year 1 establishes and validates EgoLWM and GGPO on egocentric procedural tasks (Ego-Exo4D [8] and the preliminary proposal's procedural-video evaluation suite) and demonstrates the integrated system end-to-end on the Ego Humanoid Manipulation Benchmark [21]. Year 2 extends both components to digital and UI environments, targeting the Galaxy AI on-device assistant scenario: screen recordings replace egocentric video, cursor trajectories replace ego-motion, and UI element descriptions replace action narrations, seamlessly bridging physical and digital task execution. Year 2 also delivers an open-ended procedural-planning evaluation suite generalizing the closed-loop evaluation protocol established in Year 1. Year 3 focuses on characterizing the deployment properties of the integrated system. The latent-predictive architecture is well-suited to on-device inference compared to pixel-generative world models. Year 3 will measure inference latency, memory footprint, and robustness to quantization across representative hardware targets, and study integration patterns with Samsung's existing on-device perception stack [26], in collaboration with Samsung partners who own the on-device infrastructure.

Research plan and technical approach
Data. Ego-Exo4D [8] (1,300 hours, 8 activity domains, 6DOF poses, narrations, proficiency labels, ~14M frames of manual and pseudo-GT hand annotations); HowTo100M [5] (136M clips with ASR for GGPO weak supervision); CaptainCook4D [15] and HT-Step [12] (clean procedural annotations for GGPO); EgoPER [16], ExAct [17], COIN [18], CrossTask [19], and Ego4D-v2 [20] for downstream EgoLWM evaluation; Ego Humanoid Manipulation Benchmark [21] for the end-to-end embodied demonstration.

EgoLWM: an action-aligned egocentric world model. Predicting the near-future evolution of an egocentric scene is hard because many continuations are consistent with the observed context, and a model that hedges across them gets the prediction wrong either way. The conventional answer is to condition on a language description of the wearer's intent ("the cook whisks the eggs"), and procedural planners built on this idea have made steady progress. But language descriptions are coarse: there are many ways to whisk the eggs, and the difference between a smooth incorporation and an aggressive splatter is not in the description but in how the body moves. The same holds across most egocentric activities: language is shared between successful and failed attempts, and what separates them is the motion itself. A strong such signal is the camera trajectory, the path that the wearer's motion carves through space, recorded directly by the head-mounted camera. Trajectory is physically grounded and fine-grained: the same motion that produces the action also produces the trajectory, so the trajectory carries the action's structure in a form that distinguishes between visually similar but semantically different futures. For the prediction setting, however, the relevant signal is future trajectory, which is not available at test time. A usable model must therefore predict plausible future trajectories from observation and condition action prediction on them.

Preliminary diagnostics: V-JEPA latent space is not a planning space. A natural first question is whether the standard recipe (frozen V-JEPA-style encoder, latent predictor, simple distance-based CEM scoring) transfers to long-horizon egocentric procedural planning. Our group has investigated this on Ego-Exo4D atomic-action segments, using a frozen V-JEPA 2 [2] backbone with a learned trajectory-conditioned predictor as the diagnostic regime. The findings are negative in an instructive way. Across horizons H=3 to H=8, we compare three settings of an identical latent predictor: No-Traj (no trajectory input), Oracle (ground-truth future trajectory), and CEM (search over trajectories scored by simple distance from the predicted final latent to the goal latent). Figure

3 shows that Oracle substantially outperforms No-Traj at every horizon, confirming the trajectory signal is informative; CEM, however, underperforms even No-Traj.

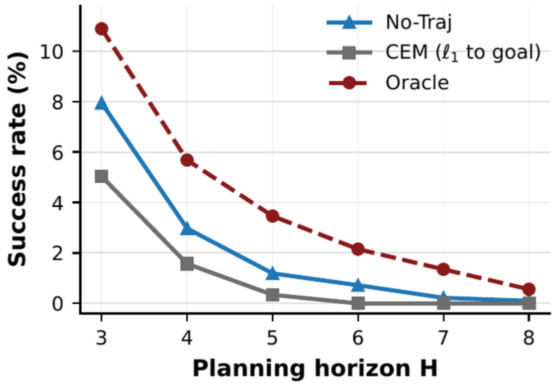

Figure 3: Latent-space CEM with simple distance-based scoring underperforms a no-trajectory baseline at every horizon. The Oracle setting shows the trajectory signal is informative; CEM cannot recover it from the V-JEPA latent geometry.

A direct probe of latent geometry explains why (Table 1). Clips from the same continuous recording are substantially closer in V-JEPA space than clips of the same action drawn from different recordings: V-JEPA latents organize by visual continuity, not by action semantics. Because distance in this space does not track action identity, scoring candidate trajectories by simple distance to a goal latent is structurally inappropriate for action-level planning. The implication is structural, not incidental: plugging in a stronger self-supervised encoder (LeWM [22], V-JEPA 2 [2]) does not change the diagnosis, since both optimize for visual continuity and inherit the same property.

Pair type	L1	Cos
Adjacent segments, same recording	0.171	0.975
Far segments, same recording	0.205	0.964
Same action, different recording	0.256	0.944
Random cross-recording pair	0.369	0.886

Table 1: V-JEPA latent geometry on Ego-Exo4D atomic-action segments. Same-recording proximity exceeds same-action cross-recording proximity, so raw latent distance does not function as a clean action metric.

Preliminary diagnostics: trajectory dominates language as a per-step conditioning signal. A separate study characterizes which input signal carries the action-discriminative content. Table 2 reports validation L1 for short-range future-latent prediction on Ego-Exo4D (H=1, predicting one action ahead) under four training regimes (no conditioning, language, future trajectory, and both) with the ground-truth future trajectory provided as oracle. Two shuffled-input ablations replace each input at validation time with an unrelated example. Under oracle trajectory, the trajectory channel carries essentially all the disambiguating content; the language channel is largely content-agnostic. This establishes the conditioning landscape under oracle access. Since oracle future trajectory is not available at inference, a usable model must predict plausible future trajectories from observation and condition on the predicted trajectory.

Conditioning	none	text	trajectory	text+traj	shuf. text	shuf. traj
Best val L1	0.4773	0.4761	0.4572	**0.4569**	0.4575	0.4964
Gain over none	—	-0.001	-0.020	-0.020	-0.020	+0.019

Table 2: Future-latent prediction on Ego-Exo4D at H=1, under oracle future trajectory. Trajectory cuts validation L1 roughly 20x more than language. Shuffling trajectory at validation degrades L1 by +0.039; shuffling language degrades it by only +0.0006.

Design. EgoLWM responds to these diagnostics with three coupled design choices. The proposed Year-1 work moves beyond the frozen V-JEPA 2 + trajectory-conditioned predictor diagnostic regime to an end-to-end training framework that addresses the underlying latent-geometry limitation. (1) *Action-aligned latent target.* We propose to train EgoLWM end-to-end to predict in an action-aligned semantic latent space whose geometry is organized by action semantics. Predictor outputs are aligned with embeddings of language-described atomic actions, so that nearby points correspond to similar actions rather than to visually similar futures. Action labels come from large-scale egocentric corpora (Ego-Exo4D's 148K labeled atomic segments [8], with additional coverage from Ego4D, EPIC-Kitchens-100, EgoPER [16], HoloAssist); the inference interface does not require language labels. (2) *Trajectory as the per-step disambiguating signal, language as training-time anchor.* EgoLWM conditions on a predicted future 6DOF ego-motion trajectory as the per-step input. The conditioning is asymmetric as in the preliminary proposal: ego-motion is injected globally via Adaptive Layer Normalization, while language acts as the structuring signal for the action-aligned readout space at training time rather than as a per-step inference input. The Year-1 work develops the trajectory predictor that closes the gap between oracle and inference, drawing on the PI's group's prior work on 6DOF egocentric pose estimation [1], which demonstrated that camera trajectories estimated from RGB video alone (via pi3) perform competitively with sensor-derived Aria trajectories for downstream egocentric prediction. This is critical for Samsung-relevant deployment: EgoLWM operates on trajectories estimated from the camera feed alone and does not require specialized IMU or SLAM hardware, making it deployable across the full Samsung device range from phones to AR glasses. A trajectory dropout mechanism, retained from the preliminary proposal, enables graceful degradation when trajectory estimation is unreliable. (3) *Hand pose as a benchmark-aligned readout, with dual training and inference role.* For the Ego Humanoid Manipulation Benchmark, whose action space is MANO hand parameters plus wrist pose, EgoLWM is trained with 3D hand pose as a secondary supervised target alongside the action embedding. The 3D hand-pose ground truth comes from Ego-Exo4D's ~14M frames of manual and pseudo-GT hand annotations. The hand-pose readout serves a dual purpose: at training time it shapes the latent geometry; at inference time the same readout head decodes predicted latents into the MANO action space required by the benchmark.

Training framework: extending LeWM. The proposed EgoLWM training framework extends LeWM's [22] stable training recipe. LeWM's two-term objective combines predicting the next latent embedding with a regularizer (SIGReg) that pushes the latents toward an isotropic Gaussian distribution, preventing collapse. The proposed extension replaces the isotropic-Gaussian target with an action-aligned target distribution defined by language-grounded action embeddings. This is a non-trivial technical move: the regularizer LeWM relies on is constructed specifically for the isotropic-Gaussian setting, and the proposed work will design and empirically validate a hybrid regularizer that retains the anti-collapse behavior of SIGReg on directions orthogonal to the action manifold while pulling latents toward the manifold along action-aligned directions. The design preserves LeWM's training simplicity while re-targeting the latent geometry from visual to action structure. The action-aligned target requires action supervision during pretraining, in contrast to LeWM's fully self-supervised regime; the same grounding mechanism that supplies GGPO's RL signal can additionally serve as a pseudo-labeling mechanism for EgoLWM pretraining, providing a unified verification primitive shared across both components.

GGPO: Grounding-Guided Planning Policy. The planner design follows the GGPO framework introduced in our preliminary proposal: a language-conditioned policy that generates multi-step plans as sequences of language-described subgoals, trained with Group Relative Policy Optimization (GRPO) [7] under a grounding-based reward. A generated subgoal sequence earns reward in proportion to how well it grounds in real ASR narrations from a large unlabeled video corpus (HowToCaption, HowTo100M-scale [5]). The key shift is that grounding is treated as a verification problem rather than a labeling problem: a candidate plan needs to be consistent with *some* expert execution observed in the wild, not match a single annotated trajectory. Multiple valid execution orderings ("add eggs then flour" vs. "add flour then eggs") both ground successfully, so the policy learns a distribution over valid plans rather than collapsing onto a single trajectory, preserving the plan diversity that downstream replanning requires. Training combines clean-label supervision on HT-Step [12] and Ego4D Goal-Step [13] with weakly-labeled HowTo100M-scale [5] data through the grounding-based reward, scaling the policy across data regimes. Our group has conducted initial proof-of-concept experiments validating the feasibility of grounding-based reward for planner training.

Integrated Anticipate–Simulate–Replan Loop. EgoLWM and GGPO are coupled into a closed-loop reasoning agent through three operations at each decision step. *Anticipate:* GGPO proposes a small set of candidate subgoal sequences for the current goal. *Simulate:* EgoLWM rolls each candidate forward in its action-aligned semantic latent space; because the prediction is in the same space where the planner's subgoals are anchored, the simulated trajectory can be evaluated for plan consistency without pixel-level decoding. *Replan:* the agent executes the candidate whose simulated rollout best matches its planned subgoal sequence; after execution, the observed latent trajectory is compared to the predicted one, and a divergence exceeding a calibrated threshold (a surprise signal) triggers GGPO to revise the remaining plan. Neither LeWM [22], AdaWorld [23], nor EgoAgent [25] implements this closed-loop revise-on-surprise regime, which recent empirical work [24] identifies as the bottleneck for current agentic systems.

End-to-End Demonstration Plan. We propose the Ego Humanoid Manipulation Benchmark [21] as the end-to-end embodied demonstration. We propose manipulation rather than navigation because the proposed EgoLWM architecture (action-aligned latents shaped by hand-object interaction supervision) and our group's published research line in egocentric procedural activity directly target hand-object task structure, the natural Samsung-relevant application domain. The benchmark is built on NVIDIA Isaac Lab, features the Unitree H1 humanoid with two Inspire dexterous hands, and provides 12 tabletop bimanual manipulation tasks: 7 short-horizon atomic actions (Push-Box, Flip-Mug, Pour-Balls, Close-Drawer, Open-Drawer, Open-Laptop, Stack-Can) and 5 long-horizon multi-stage skills (Sort-Cans, Insert-Cans, Unload-Cans, Insert-And-Unload-Cans, Stack-Can-Into-Drawer). The observation space provides egocentric RGB-D; evaluation is across 5 rooms and 5 tables in seen and unseen visual configurations, with 100 demonstrations per task. Its action space (3D wrist pose plus MANO hand parameters) aligns directly with our EgoLWM hand-pose supervision and with Ego-Exo4D's ~14M-frame hand annotations. We report Success Rate (fraction of trials completed within a time limit) and Progress (fraction of subgoals completed before failure), separately on seen and unseen visual backgrounds. The primary baseline is EgoVLA [21] itself; we also report the published no-pretraining baseline. For long-horizon tasks we additionally report a reactive variant of our own system (GGPO planner alone producing a single subgoal sequence at episode start, with EgoLWM executing each subgoal by decoding into MANO actions via the hand-pose readout, and no mid-execution replanning). The reactive-versus-closed-loop gap directly quantifies the value of integrating the world model into the decision loop. We expect that on short-horizon atomic tasks our system and EgoVLA perform comparably, and that on long-horizon multi-stage tasks the full closed-loop system outperforms both EgoVLA and the reactive variant, with the gap most pronounced on unseen visual backgrounds. If the closed-loop-versus-reactive gap is smaller than projected, that itself is an informative finding: it would indicate that the principal value of action-aligned latent prediction lies in plan quality at episode start rather than in mid-execution replanning, revising but not refuting the underlying methodological claim. The preliminary proposal's procedural-video benchmarks (EgoPER, ExAct, COIN, CrossTask, Ego4D-v2) are retained as the world-model diagnostic surface in its native domain.

Expected outcomes and results

Deliverables: 6-month (Feb 2027): EgoLWM training codebase under the LeWM-extended hybrid regularizer, with pretrained checkpoints released under a permissive open-source license; ablation results on procedural-video evaluation (ExAct [17] proficiency) isolating the contribution of the hybrid regularizer versus visual-continuity baselines, end-to-end versus frozen encoder, and AdaLN versus additive trajectory injection; trajectory-predictor codebase that closes the oracle-inference gap; one publication submission to a top-tier venue (CVPR, NeurIPS, ECCV, or ICLR). 12-month (Aug 2027): full EgoLWM + GGPO integrated system codebase with the Anticipate–Simulate–Replan loop implemented; results on the Ego Humanoid Manipulation Benchmark [21] comparing the integrated system against EgoVLA, no-pretraining, and the reactive variant, across seen and unseen visual backgrounds, on both short-horizon and long-horizon task splits; results on procedural planning and long-horizon anticipation benchmarks (COIN [18], CrossTask [19], Ego4D-v2 [20]); public release of a closed-loop evaluation protocol for Anticipate–Simulate–Replan agents, including the reactive-versus-closed-loop comparison harness and surprise-signal calibration methodology; at least one additional top-tier publication submission on the closed-loop integration; technical report including the diagnostic studies, design rationale, and full empirical evaluation, released alongside the code.

Scientific impact: The integrated program contributes (i) the first action-aligned extension of stable JEPA-style world-model training, with empirical characterization of the hybrid regularizer required to depart from isotropic priors; (ii) empirical characterization of when closed-loop Anticipate–Simulate–Replan delivers measurable benefit over reactive policies on long-horizon manipulation, addressing the foresight-integration bottleneck identified in recent empirical work [24] on agentic systems; (iii) a unified architectural substrate spanning physical and digital task domains; and (iv) deployment-readiness analyses for on-device inference in collaboration with Samsung partners.

Team composition

Lorenzo Torresani (PI) — Professor of Computer Sciences and President Joseph E. Aoun Chair, Northeastern University. PhD in Computer Science, Stanford University, 2005. Pioneer of influential architectures for image and video analysis including C3D (12,000+ citations) and TimeSformer. Co-creator of Ego-Exo4D (1,500+ registered users worldwide). Former Research Director at Meta FAIR, leading a large team in multimodal research; previously held tenured faculty position at Dartmouth College and research roles at Microsoft Research, Like.com (acquired by Google), and Digital Persona. Author of 90+ peer-reviewed papers with h-index 69; holds 11 U.S. patents. Awards include NSF CAREER Award, CVPR Best Student Paper, Google Faculty Research Award, three Facebook Faculty Research Awards, Fulbright U.S. Scholar Award, and selection as one of 16 worldwide finalists for the Microsoft Research Faculty Fellowship.

PhD student in computer vision — The graduate research assistant for this project will be recruited through Northeastern University's PhD admissions process for Fall 2026 enrollment, aligned with the project start date of September 1, 2026. The selected student will have a demonstrated background in computer vision and machine learning, ideally with prior exposure to robot simulation environments. Initial proof-of-concept experiments on GGPO have already been conducted, providing a strong foundation. The student will focus on EgoLWM architecture implementation, Ego-Exo4D training pipeline, and procedural-video benchmark evaluation in Q1-Q2; transition to full GGPO scaling, alignment method development, and weak-supervision training on HowTo100M in Q2-Q3; develop the Isaac Lab and Unitree H1 simulation pipeline and run the Ego Humanoid Manipulation Benchmark evaluation in Q3; and integrate the closed-loop Anticipate–Simulate–Replan system for the end-to-end embodied demonstration in Q4.

PART 3: Appendices (Resources & Others)

- **Figures**
- **References**
- **Budget and budget justification**
- **Facilities and equipment description**
- **Other Relevant Information**
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Figures

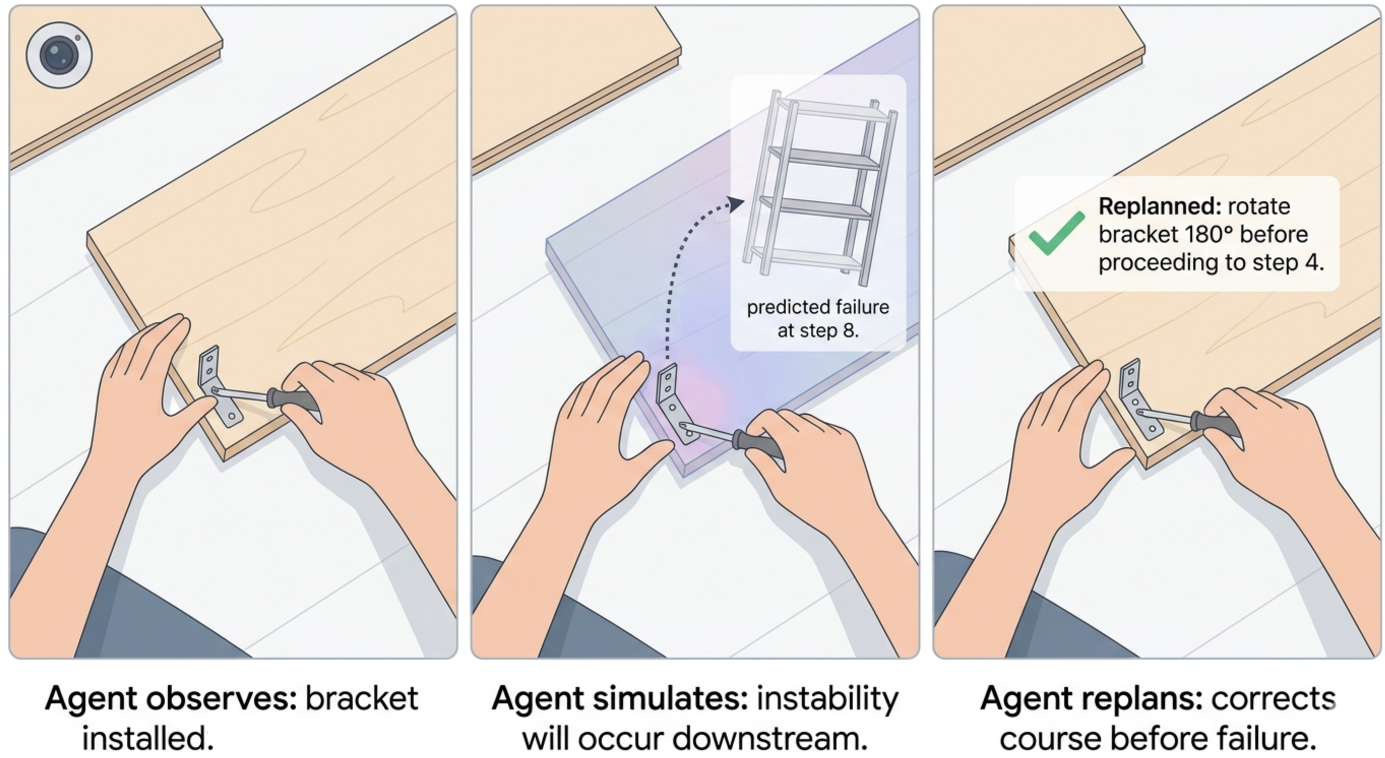

*Figure 1: **Teaser.** Three-panel illustration showing an agent observing a bracket being installed (left), simulating in latent semantic space that instability will occur downstream (center), and issuing a replanning instruction before failure occurs (right).*

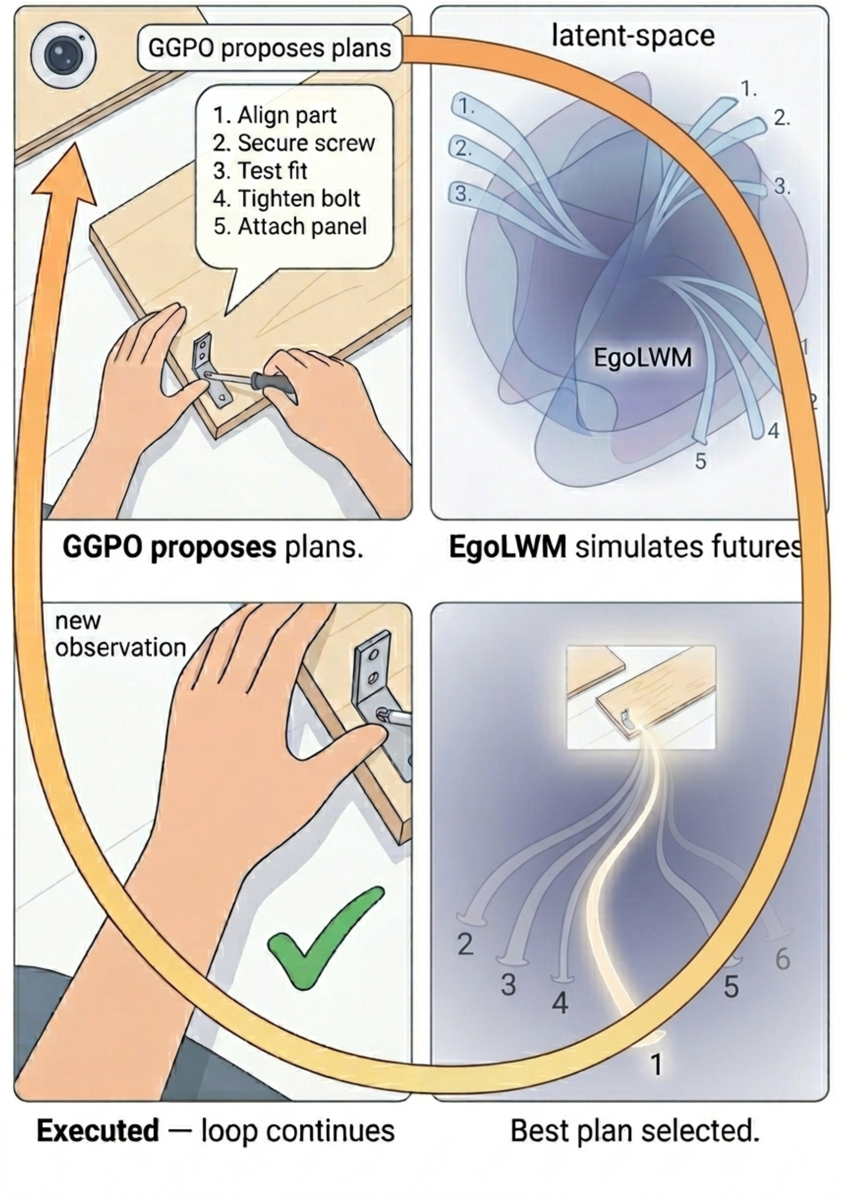

*Figure 2: **Integrated system architecture.** GGPO generates K candidate step sequences, each serving as a language conditioning signal for EgoLWM. EgoLWM simulates the future semantic latent trajectories implied by each candidate and scores them against a goal latent. The highest-scoring plan is selected and executed; new egocentric observations feed back into GGPO, closing the loop. This tight coupling, where planning informs simulation and simulation informs planning, enables the system to anticipate failures before they occur rather than react to them after the fact.*

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Budget and budget justification

Budget assumes a 5% escalation in direct costs per year.

Be advised that for the purposes of determining effort as it relates to the NSF salary limit, NU defines a “year” as September 1 through August 31.

SENIOR PERSONNEL

The Principal Investigator, Prof. Torresani, will direct all aspects of the research project including supervision of the graduate student and the design and execution of all experiments, algorithms, and tools, as well as dissemination of results. 1.6 months of summer support is budgeted for Prof. Torresani in the project year.

Total cost for all project years: $61,107

OTHER PERSONNEL

1 Graduate student research assistant (RAs)

One RA will be supported for 12 calendar month, 20 hours a week for total effort of 6 person months, for the project. The RA will work on all aspects of the proposed research under the PI's supervision. Responsibilities include implementing the EgoLWM architecture and training pipeline on Ego-Exo4D, developing and scaling the GGPO reinforcement learning framework on HowTo100M-scale data, conducting benchmark evaluations, contributing to the integrated system demonstration, and disseminating results through scholarly publications and conference presentations.

RA support consists of salary, fringe benefits, and tuition remission.

Total salary cost for all years: $53,300

FRINGE BENEFITS

The fringe benefit rate for full-time employees applied to Senior Personnel is 25.8%.

Total cost for all project years: $15,766

The fringe benefit rate for part-time employees and students applied to the Graduate Research Assistant is 7.65%. Total cost for all years: $4,077

Total cost for all years: $19,843

OTHER DIRECT COSTS

Tuition: Tuition remission budget is 50% of total cost of the normal course load of 16 credits per student. Expense is calculated, in project year one, as cost per credit $1,969 multiplied by eight credits per academic year per student. Year 1 cost for one student is $15,750 with a 5% escalation in subsequent years.

Total tuition cost for all years: $15,750.

INDIRECT COSTS

N/A

Total project cost: $150,000.

Facilities and equipment description

The Khoury College of Computer Sciences at Northeastern University provides exclusive computing resources for faculty, research staff, and graduate students. These facilities are interconnected via a layer-3 switched Ethernet gigabit network. Northeastern’s ITS Research Computing (ITSRC) delivers high-performance research computing resources to all university researchers and manages Northeastern’s participation in the Massachusetts Green High Performance Computing Center. Resources include centralized HPC clusters, storage, visualization capabilities, and specialized software. The shared Discovery cluster provides secure access to over 20,000 Intel compute cores and 200+ high-performance GPUs. The PI's research group has priority access to two dedicated high-performance GPU servers (Exxact TensorEX TS4, 4U rackmount), each equipped with 8x NVIDIA H200 NVL 141GB GPUs, dual AMD EPYC 9354 32-core processors, and 1.15TB DDR5 ECC registered memory. These servers are managed by ITSRC and integrated into the Discovery cluster infrastructure.

Other Relevant Information

External Funding. The proposed research is not currently supported by any external funding. The PI has no active or pending awards that overlap in scope with this proposal.

Background IP. The datasets and codebases underpinning this work (Ego-Exo4D, Video ReCap, BIMBA, HT-Step, Ego4D Goal-Step) are publicly released research artifacts. No proprietary or encumbered intellectual property is incorporated into the proposed research. Any IP generated under this project will be governed by the terms of the START Research Agreement.

CV of PI **Lorenzo Torresani**
Khoury College of Computer Sciences, Northeastern University, Boston, MA
L.Torresani@northeastern.edu · +1 (650) 690-0070 · https://ltorresa.github.io

Current Position: Professor & President Joseph E. Aoun Chair, Khoury College of Computer Sciences, Northeastern University (2025–present)

Education: Ph.D., Computer Science, Stanford University, 2005

Selected Prior Experience

- **Research Manager → Research Director**, Fundamental AI Research (FAIR), Meta (2021–2025) — Project lead, GenAI foundation model for Meta AI coach, video understanding models (TimeSformer)
- **Professor (tenured)**, Dartmouth College (2009–2021)
- **Research roles**: Microsoft Research (Cambridge, UK), Like.com (acquired by Google)

Research Impact: 90+ peer-reviewed publications · **h-index: 69** · 11 U.S. patents

- **C3D** (Learning Spatiotemporal Features with 3D Convolutional Networks, ICCV 2015): **12,000+ citations**; foundational architecture for video understanding
- **TimeSformer** (Is Space-Time Attention All You Need for Video Understanding?, ICML 2021): **3,700+ citations**; first convolution-free video transformer architecture

Selected Relevant Publications

- **Ego-Exo4D** (CVPR 2024, oral, <1% accept rate) — large-scale egocentric/exocentric dataset with 6DOF poses, narrations, proficiency labels; 1,500+ registered users worldwide
- **PerceptionLM** (NeurIPS 2025, spotlight) — open-access models for detailed visual understanding
- **BIMBA** (CVPR 2025) — selective-scan compression for long-range video QA; winner, EgoSchema Challenge at CVPR 2025
- **Video ReCap** (CVPR 2024) — recursive captioning of hour-long videos
- **HT-Step** (NeurIPS 2023) — aligning instructional articles with how-to videos
- **Ego4D Goal-Step** (NeurIPS 2023, spotlight) — hierarchical understanding of procedural activities
- **Step Differences in Instructional Video** (CVPR 2024)
- **Ego4D** (CVPR 2022, oral; TPAMI 2024) — 3,000 hours of egocentric video from around the world

Selected Awards

- Winner, EgoSchema Challenge, Ego4D, CVPR 2025
- 3× Distinguished Paper Award, EgoVis, CVPR 2025
- NSF CAREER Award (2010)
- Google Faculty Research Award (2014)
- 3× Facebook Faculty Research Award (2015, 2016, 2018)
- Fulbright U.S. Scholar Award (2017)
- CVPR Best Student Paper Award (2001)
- Microsoft Research Faculty Fellowship Finalist — selected as one of 16 most promising CS researchers worldwide (2011)

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